Trinh Duc Hiep

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EDUCATION

- **VNU University of Engineering and Technology** — Hanoi
  Bachelor of Science in Computer Science (Honors program); GPA: 3.53 — *Sep. 2021 - Jul. 2025*
- **Thai Binh Gifted High School** — Thai Binh
  Specialized in Mathematics — *Sep. 2018 - July. 2021*

PUBLICATIONS

- **Towards An Efficient Multimodal Approach for Long Video Summarization** — FJCAI 2026
  Trung-Hieu Cao, Duc-Hiep Trinh*, Tran-Binh Dang, Duc-Trong Le and Phi-Diep Bui* — *Vietnam*

RESEARCH EXPERIENCES

- **Legal Texture Entailment [Link]** — Hanoi
  Researcher — *July 2023 - December 2023*
  - Researched Legal Texture Entailment using NLP, Legal Informatics to model complex legal texts for semantic entailment.
  - Developed and fine-tuned transformer-based ML models, improving entailment prediction accuracy by 15% over baselines.
  - Collaborated with senior researchers to analyze results, interpret metrics, and co-author a technical report with future research proposals.
- **Enhancing Video Summarization Techniques with Advanced Language Models** — Hanoi
  Researcher — *December 2023 - March 2025*
  - Built and maintained a sports database for top five European football leagues, enabling detailed analysis of match results and player stats.
  - Designed a workflow for automated video summarization using BERT and GPT models to generate concise football match summaries.
  - Enhanced video summarization with computer vision and NLP algorithms, improving accuracy by 20%.
  - Tested and validated summarization algorithms with quantitative metrics to ensure high-quality outputs.
- **A Multimodal Method for Identifying Foul Situations in Football Matches [Link]** — Hanoi
  Researcher — *March 2025 - July 2025*
  - Spearheaded the development of a multimodal foul detection framework, combining computer vision and temporal data processing to accurately identify foul situations in football matches with high precision.
  - Designed and implemented a novel data augmentation technique to address class imbalance in foul detection datasets.
  - Achieved a notable improvement in foul detection accuracy, boosting model performance by 2 % through the integration of multimodal feature fusion and fine-tuning of deep learning models.

WORK EXPERIENCES

- **MotionsCloud** — Hanoi
  AI Engineer — *September 2025 - present*
  - Enhanced the InternVL model for the Singapore Airlines (SIA) project (a computer vision initiative focused on classifying luggage damage) achieving a Macro F1 score of 0.908 compared to 0.8 of the baseline model through data augmentation and preprocessing optimizations.
  - Curated a dataset with 1.8k case folders and 12k+ images for key damage types (body crack, dent, wheel, rubber), expanded annotations for non-damage components (handles, lock, zipper), and cleaned/integrated 5.7k additional images to enhance model robustness.
  - Developed confidence in utilizing terminal commands for system management, Docker container orchestration, and general workflow automation.

ACADEMIC ACHIEVEMENTS

- **Top 5 of Vietnamese Language and Speech Processing**
  Issued by Vietnam Institute for Advanced Study in Mathematics(VIASM)
- **Consolation Prize for Scientific Research Conference for Undergraduates 2024**
  Issued by Faculty of Information and Technology of University of Engineering and Technology - Vietnam National University
- **First Prize for Scientific Research Conference for Undergraduates 2025 (Faculty Level)**
  Issued by Faculty of Information and Technology of University of Engineering and Technology - Vietnam National University
- **Third Prize for Scientific Research Conference for Undergraduates 2025 (University Level)**
  Issued by University of Engineering and Technology - Vietnam National University

SKILLS

- **Technical Skills**
  Programming
  - Language: Python, C++
  - Framework: Deep Learning, Machine Learning, Math Modeling.
- **Soft Skills**
  Language
  - English: IELTS 6.5
  - Vietnamese: Native or bilingual proficiency